Seattle Infinity Math Circle TST **September 20th, 2026**

Computational Round **75 minutes**

---

1. Three ants, Christopher, Kevin, and Raymond, crawl at the same speed on the Cartesian plane. Christopher starts at $(0, 0)$ and each second moves either one unit up or right; Kevin starts at $(2, 8)$ and moves down or right; Raymond starts at $(9, 1)$ and moves up or left. If all three eventually meet at the same point without any two meeting earlier, how many possible sets of paths are there?

   Solution. If the ants meet after t seconds at (x, y), then

   $$x + y = t, \qquad y - x = 6 - t, \qquad y - x = t - 8,$$

   so $t = 7$ and the meeting point is $(4, 3)$.

   Ignoring earlier meetings, the number of path sets is

   $$\binom{7}{3}\binom{7}{2}^2 = 35 \cdot 21^2 = 15435.$$

   We subtract paths where two ants meet early. For Christopher and Kevin, an early meeting can only occur at $t = 5$ or 6, and meeting at $t = 5$ forces them to meet again at $t = 6$. Thus they meet early iff both are at $(3, 3)$ after 6 steps. This gives

   $$\binom{6}{3}\binom{6}{1}\binom{7}{2} = 20 \cdot 6 \cdot 21 = 2520.$$

   Similarly, Christopher and Raymond meet early iff both are at $(4, 2)$ after 6 steps, giving

   $$\binom{6}{2}\binom{6}{1}\binom{7}{2} = 15 \cdot 6 \cdot 21 = 1890.$$

   Kevin and Raymond cannot meet before $t = 7$, and the two bad cases above are disjoint. Hence, our answer is

   $$15435 - 2520 - 1890 = \boxed{11025}.$$

2. Let p_1, p_2, $\ldots$, p_8 be distinct primes such that for all i,

$$\begin{aligned} p_i^2 - p_i + 1 &\equiv 0 \pmod 7 \\ p_1 + p_2 + \cdots + p_8 &\equiv 2 \pmod 7. \end{aligned}$$

How many positive divisors of $N = p_1 p_2 \ldots p_8$ are congruent to 1 mod 7?

Solution. Note that for $p_i^2 - p_i + 1 \equiv 0 \bmod 7$, we require $p_i \equiv 3, 5$. Thus, among $p_1, \ldots, p_8$, 3 are 5 mod 7 and 5 are 3 mod 7, by the second equation. Observe that $3 \cdot 5 \equiv 1 \bmod 7$, so we must select the same number of primes of each type to construct our divisor.

This gives

$$\binom{5}{0}\binom{3}{0} + \binom{5}{1}\binom{3}{1} + \binom{5}{2}\binom{3}{2} + \binom{5}{3}\binom{3}{3} = \boxed{56}.$$

3. Let $\theta = 2\pi/2027$. Define

$$A = \sum_{n=0}^{675} \cos(3n\theta).$$

Compute

$$\frac{1}{A} - \frac{1}{\cos\theta}.$$

Solution. Letting $\omega = \exp(i\theta)$, complex nums gives

$$\begin{aligned}
&\sum_{n=0}^{675} \cos(3nd) \\
&= \operatorname{Re}\left(\sum_{n=0}^{675} \omega^{3n}\right) \\
&= \operatorname{Re}\left(\frac{\omega^{3\cdot 676} - 1}{\omega^3 - 1}\right) \\
&= \operatorname{Re}\left(\frac{\omega - 1}{\omega^3 - 1}\right) \\
&= \operatorname{Re}\left(\frac{1}{\omega^2 + \omega + 1}\right) \\
&= \operatorname{Re}\left(\frac{1}{\omega}\right) \frac{1}{\omega + 1 + 1/\omega} \\
&= \frac{\cos(\theta)}{1 + 2\cos(\theta)} \\
&= \frac{1}{\sec\theta + 2}
\end{aligned}$$

Thus the desired answer is $\boxed{2}$.

4. Let $\triangle ABC$ be an acute triangle with orthocenter H and circumcenter O. Let the midpoint of $\overline{BC}$ be M. If $MO = 6$, $MH = 7$, and $MA = 19$, compute BC.

Solution. We have $AH = 2OM = 12$. Let the midpoint of AH be N, and recall that $OMHN$ and $OMNA$ are parallelograms. Using Stewart's on triangle AHM we get

$$NM^2 \cdot 12 + 6 \cdot 6 \cdot 12 = 19^2 \cdot 6 + 7^2 \cdot 6$$
$$NM^2 \cdot 2 + 6 \cdot 12 = 19^2 + 7^2$$
$$NM = 13$$

Now note that NM is simply the circumradius of ABC. By Pythag, $BC = 2BM = 2\sqrt{13^2 - 6^2} = \boxed{2\sqrt{133}}$.

5. Kevin plays a game on the number line. Each second, if at position n, he gains n points, then moves to $n-1$ or $n+1$ with equal probability. The game ends when Kevin reaches 0 or 15.

   If he starts at position 6 with 0 points, what is Kevin's expected final score?

   Solution. Let $E(x)$ be the expected number of points if starting at position x. Note that

$$E(x) = \frac{1}{2}E(x-1) + \frac{1}{2}E(x+1) + x,$$

   while $E(0) = 0$ and $E(15) = 0$. As a result, we have that

$$E(x-1) - 2E(x) + E(x+1) = -2x,$$

   or that the second difference of E is linear. As a result, we guess that

$$E(x) = ax^3 + bx^2 + cx + d.$$

   We end up solving that

$$E(x) = \frac{x(15-x)(15+x)}{3},$$

   so $E(6) = \boxed{378}$.

6. Christopher constructs a random sequence of real numbers $a_1, a_2, a_3, \ldots$ as follows. Let $a_1 = \frac{3}{4}$. For each $n \geq 2$, he independently chooses a_n to be $\frac{3}{4}|a_{n-1}|$ with probability $\frac{2}{3}$ and $-a_{n-1}^2$ with probability $\frac{1}{3}$. Find the expected value of $\sum_{n=1}^{\infty} a_n$.

Solution. Note that

$$|a_1| + |a_2| + \cdots \leq \frac{3}{4} + \left(\frac{3}{4}\right)^2 + \cdots = 3,$$

so $a_1 + a_2 + \cdots$ is absolutely convergent.

We prove that for all m, the expected value of

$$a_1 + a_2 + \cdots + a_m + |a_m|$$

is $\frac{3}{2}$. Taking $m \to \infty$, the expected value of $|a_m|$ approaches 0, so the answer is $\boxed{\frac{3}{2}}$.

We use linearity of expectation. Let A_n denote the multiset of all possible values of a_n. Note that $a_{n+1} + |a_{n+1}|$ is 0 with probability $\frac{1}{3}$ and $\frac{3}{2}|a_n|$ with probability $\frac{2}{3}$. Thus, the expected value of $a_{n+1} + |a_{n+1}|$ over all $a_{n+1} \in A_{n+1}$ is the expected value of $|a_n|$ over all $a_n \in A_n$. By induction, we get

$$E(a_1 + a_2 + \cdots + a_m + |a_m|) = E(a_1 + a_2 + \cdots + a_{m-1} + |a_{m-1}|) = E(a_1 + |a_1|) = \frac{3}{2},$$

as desired.

7. Find the number of ordered pairs of positive integers (x, y) with $x, y \leq 506$ such that

$$x^y \equiv y^x \pmod{23}.$$

Solution. There are 22^2 pairs where x and y are both multiples of 23.

Let g be a primitive root mod 23. Each $(a, b) \in (\mathbb{Z}/22\mathbb{Z})^2$ determines a unique integer $1 \leq n \leq 506$ with $23 \nmid n$ by $n \equiv g^a \pmod{23}$ and $n \equiv b \pmod{22}$. Let

$$\begin{aligned} x &\equiv g^{a_1} \pmod{23} \\ x &\equiv b_1 \pmod{22} \\ y &\equiv g^{a_2} \pmod{23} \\ y &\equiv b_2 \pmod{22}. \end{aligned}$$

Then, $g^{a_1 b_2} \equiv g^{a_2 b_1} \pmod{23}$, so $a_1 b_2 \equiv a_2 b_1 \pmod{22}$. It suffices to find the number of $(a, b, c, d) \in (\mathbb{Z}/22\mathbb{Z})^4$ such that $ab \equiv cd \pmod{22}$. We work over $\mathbb{Z}/22\mathbb{Z}$.

Suppose ab is a unit. For a fixed unit n, there are 10 choices for a and b such that that $ab = n$. Hence, there are $10 \cdot 10^2$ solutions in this case.

Suppose ab is nonzero and $2 \mid ab$. For a fixed nonzero $2 \mid n$, there are $10 + 10 + 10$ choices for a and b such that $ab = n$. Hence, there are $10 \cdot 30^2$ solutions in this case.

Suppose $ab = 11$. There are $(10 + 10 + 1)^2$ solutions in this case.

Suppose $ab = 0$. By PIE, there are $(22^2 - 11^2 - 20^2 + 10^2)^2$ solutions in this case.

In total, there are

$$22^2 + 10^3 + 9 \cdot 10^3 + 21^2 + (22^2 - 11^2 - 20^2 + 10^2)^2 = \boxed{14894}$$

solutions.

8. A cyclic hexagon $ABCDEF$ satisfies $AB = BC = 1$, $CD = DE = \sqrt{2}$, and $EF = FA = \sqrt{5}$. Let X be a point in the interior of the circumcircle. Find the minimum possible value of $XA + XB + XC + XD + XE + XF$.

Solution. Relative to triangle BDF, DA, BE, FC are altitudes. Thus the given expression is minimized when X is the orthocenter H of BDF, and the minimum possible value is $AD+BE+CF$. Let $a = 1, b = \sqrt{2}, c = \sqrt{5}$.

It remains to compute this value. Note that the orthocenter reflected over each side of BDF gives A, C, E, thus $HB = a, HD = b, HF = c$.

Let the power of H be $2x$, and let $X = BE \cap DF, Y = DA \cap BF, Z = FE \cap BD$. Then

$HX = \frac{1}{2}HE = x/a$, and similarly $HY = x/b$, $HZ = x/c$. The desired value is thus $2(x/a + x/b + x/c) + a + b + c$.

Now, note that

$$XD^2 + XB^2 = BD^2$$
$$HD^2 - HX^2 + XB^2 = (BZ + ZD)^2$$
$$b^2 - (x/a)^2 + (a + x/a)^2 = (\sqrt{a^2 - (x/c)^2} + \sqrt{b^2 - (x/c)^2})^2$$
$$(x/c)^2 + x = \sqrt{a^2 - (x/c)^2}\sqrt{b^2 - (x/c)^2}$$
$$2x^3 + (a^2 + b^2 + c^2)x^2 = a^2b^2c^2$$

Plugging in gives

$$2x^3 + 8x^2 = 10$$
$$x^3 + 4x^2 - 5 = 0$$
$$(x - 1)(x^2 + 5x + 5)$$

From which it is clear the only solution is $x = 1$. Now the answer is just

$$\boxed{3 + 2\sqrt{2} + \frac{7\sqrt{5}}{5}}$$

9. Ian and Raymond are playing a guessing game. Ian chooses a number $a \in [0,1]$. Initially, $s = 0$ and $t = 1$. On each turn, Raymond guesses uniformly at random from $[s,t]$. If his guess exceeds a, he sets t equal to his guess; otherwise, he sets s equal to his guess. This process repeats. If $a = \frac{1}{3}$, find the expected value of Raymond's 9th guess.

Solution. We prove that the expected value of Raymond's n-th guess is

$$\frac{1 + (2^n - 2)a}{2^n},$$

which gives an answer of $\boxed{\frac{171}{512}}$.

We use induction on n, with the base case trivial. Assume the statement for $n = k$. Suppose Raymond's first guess is $a < x < 1$. All of Raymond's future guesses lies in the interval $(0, x)$. Scaling up to $(0,1)$, Raymond's goal is now to guess $\frac{a}{x}$, so the expected value of his k-th guess after the first is

$$\frac{1 + (2^k - 2)\frac{a}{x}}{2^k}.$$

Scaling back down to $(0, x)$, the expected value of Raymond's $(k+1)$-th guess is

$$\frac{x + (2^k - 2)a}{2^k},$$

given that his first guess is x. The expected value of this over all $a < x < 1$ is

$$\frac{\frac{a+1}{2} + (2^k - 2)a}{2^k} = \frac{1 + (2^{k+1} - 3)a}{2^{k+1}}.$$

Similarly, if Raymond's first guess is in $(0, a)$, the expected value of his $(k+1)$-th guess is

$$1 - \frac{1 + (2^{k+1} - 3)(1 - a)}{2^{k+1}} = \frac{2 + (2^{k+1} - 3)a}{2^{k+1}}.$$

The former occurs with probability $1 - a$, and the latter occurs with probability a, so the expected value of Raymond's $(k+1)$-th guess is

$$a \cdot \frac{2 + (2^{k+1} - 3)a}{2^{k+1}} + (1 - a) \cdot \frac{1 + (2^{k+1} - 3)a}{2^{k+1}} = \frac{1 + (2^{k+1} - 2)a}{2^{k+1}},$$

completing the induction step.

10. Points A, B, C, D, E lie on a circle in that order with $AB = AE = 7$, $BC = CD = 3$, and $AC = 8$. Let $\overline{BD}$ intersect $\overline{CE}$ at X. Compute CX/XE.

Solution. Claim: ABE is an equilateral triangle of side length 7, and $CE = 5$.

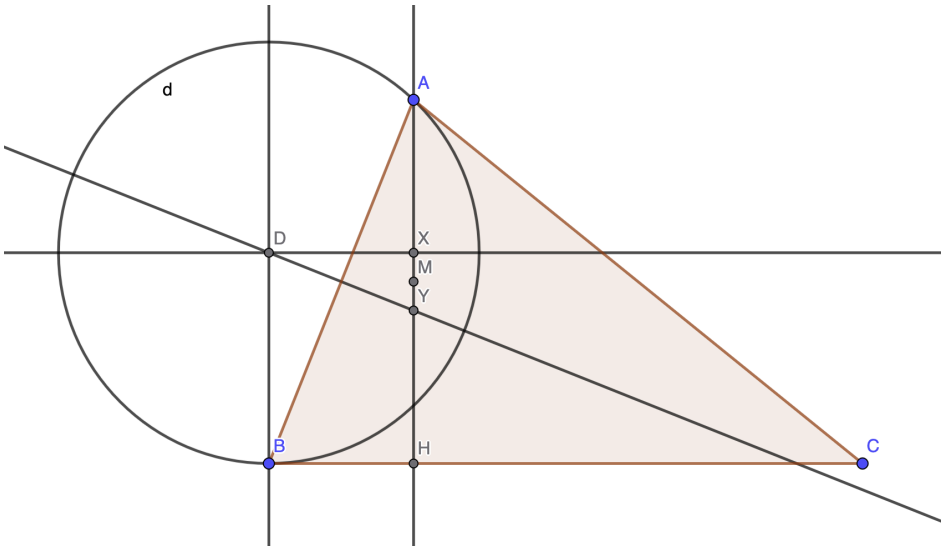
Proof: Let ABE' be an equilateral triangle of length 7 such that E' and E lie on the same side of B. Let C' satisfy $BC' = 3$, $C'E = 5$ and be on the same side of BE as C. Note that a $3 - 5 - 7$ triangle has an angle of 120°, so $ABC'E'$ is cyclic. By Ptolemy's $AC' = 8$. This implies $C = C'$, and thus $E = E'$ as desired.

Now, let Γ be the circle with center C passing through B and D and let Ω be the circle with diameter CE. The intersection of the radical axes of Γ, Ω, and $(ABCDE)$ is simply X.

Let Γ and Ω intersect at Y (the intersection point chosen does not matter). Note that since CE is a diameter of Ω, CYE is a right triangle and $YX \perp CE$. Since $CY = 3, CE = 5$, we have a 3-4-5 right triangle and $CX/XE = \boxed{\frac{9}{16}}$.

1. **[5]** Let $\triangle ABC$ be a scalene triangle. Let D be the center of the circle passing through A tangent to $\overline{BC}$ at B. Let X be the projection of D onto the A-altitude of $\triangle ABC$, and let Y be the intersection of the angle bisector of $\angle ADB$ and the A-altitude of $\triangle ABC$. Prove that the midpoint of $\overline{XY}$ is also the midpoint of the A-altitude.

   Solution.



   Let H be the foot from A to $\overline{BC}$. Our main claim is that

$$HX = BD = AD = AY.$$

   It is clear that $XDBH$ is a rectangle, as all of its angles are right by definition. Hence, $HX = BD$.

   It follows that $BD = AD$ as they both lie on the circle with center D.

   Finally, note that $\angle ADY = \angle BDY$, while since $\overline{BD} \parallel \overline{AH}$, we know that $\angle BDY = \angle AYD$, so $\angle AYD = \angle ADY$, hence $\triangle ADY$ is isosceles and $AD = AY$.

2. **[5]** Call a positive integer "d-amazing" for some digit $2 \leq d \leq 9$ if:

   - It can be expressed as a^b for some positive integers $a, b > 1$.
   - Every digit in the number is one of 0, 1, or d, and all 3 digits appear at least once.
   - The units digit is nonzero.

   Prove that there exist infinitely many d-amazing integers for each $d \in \{2, 3, 4, 5, 8\}$.

   Solution.

$$2\colon (10^n + 1)^2$$
$$3\colon (10^n + 1)^3$$
$$4\colon (10^n + 2)^2$$
$$5\colon (10^n + 1)^5$$
$$8\colon (10^n + 9)^2$$

3. **[6]** Let a, b, and c be pairwise relatively prime positive integers. Prove that

$$\gcd(a^3 + bc, b^3 + ac, c^3 + ab)$$

divides $(a^2 - 1)(b^2 - 1)(c^2 - 1)$.

Solution. Let the gcd be d, and note that a, b, c, and d are pairwise relatively prime. Since

$$\begin{aligned} a^3 &\equiv -bc \pmod{d} \\ b^3 &\equiv -ca \pmod{d} \\ c^3 &\equiv -ab \pmod{d}, \end{aligned}$$

it follows that $abc \equiv -1 \pmod{d}$. Now, we have $a^4 \equiv b^4 \equiv c^4 \equiv 1 \pmod{d}$. Let $\alpha := \nu_2(d)$. If $\alpha > 0$, then a, b, and c are odd, so $\nu_2(a^2 + 1) = 1$. Hence,

$$\nu_2(d) \leq \nu_2(a^4 - 1) = 1 + \nu_2(a^2 - 1) \leq \nu_2((a^2 - 1)(b^2 - 1)(c^2 - 1)).$$

Suppose FTSOC there exists a prime power $q \mid d$ such that $q \nmid (a^2 - 1)(b^2 - 1)(c^2 - 1)$. Since $q \mid (a^2 - 1)(a^2 + 1)$ and $\gcd(a^2 - 1, a^2 + 1) = 2$, it follows that $q \mid a^2 + 1$. Similarly, $q \mid b^2 + 1, c^2 + 1$. Thus, $a^2b^2c^2 \equiv -1 \pmod{q}$, a contradiction to $abc \equiv -1 \pmod{q}$.

4. **[7]** Find all real numbers c for which there exists a nonzero function $f\colon \mathbb{R} \to \mathbb{R}$ such that

$$f(x)^2 + cf(y)^2 = (f(x+y) + f(y))(f(x-y) + f(y))$$

holds for all real numbers x and y.

Solution. We claim the answers are $c \in \{3, 4\}$. For 3, $f(x) = 1$ works. For 4, $f(x) = x^2$ works by Sophie-Germain.

Now, assume $c \notin \{3, 4\}$. AFTSOC there exists a nonzero solution. Plugging in $x = y = 0$ gives

$$(c+1)f(0)^2 = 4f(0)^2$$

. Since $c \neq 3$ we must have $f(0) = 0$. Plugging in $x = 0$ now gives

$$cf(y)^2 = 2f(y)(f(-y) + f(y))$$

If f is not the zero function, there exists some a with $f(a) \neq 0$. Then

$$cf(a) = 2f(-a) + 2f(a) \to f(-a) = \frac{c-2}{2} f(a)$$

If $c = 2$, then $f(-a) = 0$. Plugging in $(a, -a)$ gives a contradiction, since we get $f(a)^2 = f(0)f(2a) = 0$.

Thus $c \neq 2$, so $f(-a)$ is nonzero and $(\frac{c-2}{2})^2 = 1$. Since $c \neq 4$, we must have $c = 0$.

Now take $x = y$. we have $f(x)^2 = (f(2x) + f(x))f(x) \to f(x) = 0$ or $f(2x) = 0$. AFTSOC f is nonzero, then we can find x where $f(x)$ is nonzero. Thus $f(x/2) = 0$. Then plug in $y = x/2$ to get $f(x)^2 = (f(\frac{3}{2}x) + f(\frac{1}{2}x)) \cdot 2f(\frac{1}{2}x)$. But this implies $f(x) = 0$, a contradiction. Thus f is the zero function.

In all cases, if $c \notin \{3, 4\}$, f must be the zero function.

5. **[8]** Lumpy and Champy play on the infinite square grid of lattice points, with edges joining adjacent points. Initially, only the origin is black. On Lumpy's turn, he deletes any edge. On Champy's turn, she colors black any white point adjacent to a black point by an undeleted edge. They alternate turns. The game ends when Champy has no legal move. Her score is the maximum taxicab distance between two black points, where the taxicab distance between (a, b) and (c, d) is $|a - c| + |b - d|$. Lumpy plays to minimize this score. Prove that:

   (a) If Champy moves first, Lumpy can limit Champy's score to at most 16.

   (b) If Lumpy moves first, Champy will always end with a score of at least 6.

   Solution.

   This problem is easier to visualize on an infinite grid of square cells, where Champy instead fills in cells and Lumpy places walls on edges.

   Part 1: After Champy places his first black cell, he has a $2x1$ black domino. Lumpy can imagine a 9×10 bounding square and wall off a corner (WLOG top left). Then, he can further bound the rectangle to $9x9$ or $8x10$ depending on Champy's second move. On Champy's third and fourth move, Lumpy finishes walling the corners of the rectangle. The earliest Champy can reach the boundary of the rectangle is on his fifth move. Lumpy spends his remaining moves walling off the boundary of the rectangle, and since all corners are pre-walled, Champy can never escape the rectangle.

   Part 2: Consider the board state right after the game ends. Since Champy effectively moves first (considering the starting black point) and we must have ended on a Lumpy move, they have taken the same number of turns, say t. Consider the minimum rectilinear polygon enclosing all black squares, which has perimeter p and area a. Since Champy has no move, Lumpy must have walls around the entire perimeter, so $t \geq p$. Since Champy has not escaped the area, $t \leq a \rightarrow p \leq a$. But also note that $a \leq (p/4)^2$, with equality iff the polygon is a square. Thus $p \leq (p/4)^2 \rightarrow p \geq 16 \rightarrow a \geq 16$.

   Now, AFTSOC the score of the final region is ≤ 5. Consider the minimum bounding rectangle of the black cells. If the width or height is at least 7 squares, then this provides a score of at least 6. If the width is exactly 6, then the two squares with x-coordinates differing by 5 must be on the same row. We may consider all the squares at distance at most 5 from both squares, which forms a shape with 18 squares total. Then $p \leq a \leq 18$, so the height of the bounding rectangle is at most 2 - but a 6×2 bounding rectangle cannot contain a region of area at least 16, a contradiction. The same is true of the height, thus the black cells can be bounded within a 5×5 rectangle.

   Then, consider the set of 4 corners and 4 side-midpoint-cells. We can list them out in a loop such that each adjacent pair is at taxicab distance 6, showing that at least 4 of those cells are white. We can partition the remaining 8 boundary cells into 4 pairs of 2, each at taxicab distance 6 from each other, showing that at least 4 of those cells are white. This means there are at most 25-4-4 = 17 black cells. Since the area is at least than the perimeter, the perimeter is at most 17. Since perimeter is always even, it must be exactly 16. Then since $a \leq (p/4)^2 = 16 = p \leq a$, the area is also 16. This means our region must be exactly a 4x4 grid - but this has score 6, a contradiction.

6. **[10]** Let $P(x)$ be a cubic polynomial such that $P(x) = x$ and $P(P(x)) = x$ each have exactly 3 distinct real solutions. Prove that for all positive integers n, $P^n(x) = x$ has exactly 3 distinct real solutions.

Solution. We first prove that the leading coefficient of P is positive. Suppose FTSOC it is negative, and let the distinct roots of P be $r_1 < r_2 < r_3$. Note that $P(x) - x$ is negative on the interval (r_1, r_2). Since $P(x)$ is continuous and approaches $-\infty$ when $x \to \infty$, there exists $a > r_3$ for which $r_1 < P(a) < r_2$. Hence, $P(P(a)) < P(a) < a$. However, $P(P(x)) - x$ is continuous and approaches ∞ when $x \to \infty$, so there exists $r_4 > a$ such that $P(P(r_4)) - r_4 = 0$. Now, r_1, r_2, r_3, r_4 are distinct solutions to $P(P(x))$, a contradiction. Thus, the leading coefficient of P is positive.

We use induction on n.

Assume the statement for $n-1$. Note that $P(x) - x$ is positive on $(r_1, r_2) \cup (r_3, \infty)$ and negative on $(-\infty, r_1) \cup (r_2, r_3)$. Suppose FTSOC there exists $a \neq r_1, r_2, r_3$ such that $P^n(a) = a$. Let $a_i := P^i(a)$ for all $0 \leq i \leq n-1$. By the induction hypothesis, $a_0, a_1, \ldots, a_{n-1}$ are distinct. From now on, indices will be taken cyclically. For any $x < r_1$, we have

$$P(x) < x < r_1 \implies P^2(x) < P(x) < x < r_1 \implies \cdots \implies P^n(x) < x.$$

Thus, $a_i > r_1$ for all i. Similarly, $a_i < r_3$.

Fix a positive integer $k < n$. By IVT and the induction hypothesis, if $P^k(a_i) > a_i$ and $P^k(a_j) < a_j$, then $a_i < r_2 < a_j$ or $a_j < r_2 < a_i$, since $P^k(x) = x$ has a solution between a_i and a_j. Since we cannot have $P^k(a_i) > a_i$ for all i or $P^k(a_i) < a_i$ for all i, it follows that $P^k(a_i) > a_i$ iff $a_i < r_2$ or $a_i > r_2$, with the direction of the inequality determined only by k. The analogous result exists for when $P^k(a_i) < a_i$.

Let $A := \{i \mid a_i < r_2\}$ and $B := \{i \mid a_i > r_2\}$. Since $n \geq 3$, by pigeonhole one of A or B has at least 2 elements. WLOG A has two elements i and $i + k$. If $a_i > a_{i+k}$, then

$$P^k(a_i) < a_i \implies P^k(a_{i+k}) < a_{i+k} \implies a_{i+k} > a_{i+2k}.$$

Continuing like this, we get $a_i > a_{i+nk}$, a contradiction. The case $a_i < a_{i+k}$ similarly leads to a contradiction.

Thus, $P^n(x) = x$ has exactly 3 distinct real solutions, completing the induction step.

7. **[11]** Distinct points A, B, C, A', B', and C', not all coplanar, are chosen on a sphere centered at O such that AA', BB', and CC' concur at a point $P \neq O$. Prove that line OP is perpendicular to the line formed by intersecting the planes of $\triangle ABC$ and $\triangle A'B'C'$.

Solution. Let l denote the intersection of the planes of $\triangle ABC$ and $\triangle A'B'C'$, denoted Γ_1 and Γ_2 respectively.

Claim: $D := \overline{AB} \cap \overline{A'B'}$ lies on the line, and similarly for $E := \overline{BC} \cap \overline{B'C'}$, $F := \overline{AC} \cap \overline{A'C'}$

(Note: intersections at points of infinity are allowed).

Proof: for each pair, the first line lies entirely in plane Γ_1 and the second line lies entirely in plane Γ_2. Since the two lines are coplanar and non parallel, they intersect. Their intersection lies on Γ_1 and Γ_2, thus it lies on line l.

We now use vectors with origin at the center of the original sphere. We claim $D \cdot P = E \cdot P = F \cdot P = 1$. Let the circumcircle of $ABA'B'$ be centered at O_1 with radius r_1. Point D lies on the same plane, so we may simplify the problem to $2D$. D lies on the polar line of point P, so $(D - O_1) \cdot (P - O_1) = r_1^2$ by projecting vector O_1D onto line O_1P. Additionally we know

$$(D - O_1) \cdot O_1 = (P - O_1) \cdot O_1 = 0$$

$$O_1 \cdot O_1 = 1 - r_1^2$$

since the plane is perpendicular to OO_1. We may conclude that

$$\begin{aligned} D \cdot P &= (D - O_1 + O_1) \cdot (P - O_1 + O_1) \\ &= (D - O_1) \cdot (P - O_1) + (D - O_1) \cdot O_1 + (P - O_1) \cdot O_1 + O_1 \cdot O_1 \\ &= 1 \end{aligned}$$

as desired. We symmetrically have $D \cdot P = E \cdot P = F \cdot P = 1$, thus line DEF is perpendicular to OP.

8. **[12]** Mr. Least traps $N \geq 2$ contestants in his basement. The winner receives a prize pool, which is initially empty. Each day, while at least two contestants remain, Mr. Least assigns each contestant to either the *\$1 room* or the *\$2 room*, then selects one contestant uniformly at random. If the selected contestant is in the \$1 room, he adds \$1 to the prize pool and eliminates everyone in the \$2 room; if they are in the \$2 room, he adds \$2 and eliminates everyone in the \$1 room.

   Mr. Least wants to minimize the expected prize pool. Find an optimal strategy and determine for which N there is a unique optimal strategy.

   Solution. Suppose with n contestants remaining Mr. Least puts k contestants in the \$1 room and $n-k$ contestants in the \$2 room. Let F_t be the Fibonacci numbers with $F_0 = 0$ and $F_1 = 1$. We claim that if $F_t + 1 \leq n \leq F_{t+1}$, then selecting k such that $F_{t-1} \leq k \leq F_t$ and $F_{t-2} \leq n-k \leq F_{t-1}$ results in an optimal strategy, with $n = F_t$ for some t being exactly when optimal strategy is unique immediately following. This strategy results in an expected pool of $t + 1 - \frac{F_{t+2}}{n}$.

   Let V_n be the expected final pool at the optimal strategy. Then,

$$V_n = \min_{1 \leq k \leq n-1} \left(\frac{k}{n}(1 + V_k) + \frac{n-k}{n}(2 + V_{n-k}) \right).$$

   Define $A_n = nV_n$. Then,

$$A_n = 2n + \min_{1 \leq k \leq n-1} (A_k + A_{n-k} - k).$$

   We will proceed with induction over t now, claiming that $A_n = (t+1)n - F_{t+2}$ whenever $F_t + 1 \leq n \leq F_{t+1}$. For $t = 2$ it is obvious. For inductive step, suppose it works for $t-1$. Let $B_n(k) = A_k + A_{n-k} - k$. Then,

$$A_n = 2n + \min_{1 \leq k \leq n-1} (B_n(k)).$$

   Claim 1. $B_n(k)$ is minimized exactly when both

$$F_{t-1} \leq k \leq F_t \quad \text{and} \quad F_{t-2} \leq n-k \leq F_{t-1}.$$

   Proof. Define the first difference

$$D_m = A_m - A_{m-1}.$$

   By the inductive hypothesis, if $F_r + 1 \leq m \leq F_{r+1}$, then

$$D_m = r + 1.$$

   Now, note that

$$\begin{aligned} B_n(k+1) - B_n(k) &= (A_{k+1} - A_k) - (A_{n-k} - A_{n-k-1}) - 1 \\ &= D_{k+1} - D_{n-k} - 1. \end{aligned}$$

   We now have the following cases on the values of D_{k+1} and D_{n-k}:

   - *Case 1:* $k \leq F_{t-1} - 1$. Since $n > F_t = F_{t-1} + F_{t-2}$, we know that $n - k > F_{t-2}$. Thus $D_{k+1} \leq t - 1$ while $D_{n-k} \geq t - 1$, so

$$B_n(k+1) - B_n(k) < 0.$$

   - *Case 2:* $n - k \geq F_{t-1} + 1$. Since $n \leq F_{t+1} = F_t + F_{t-1}$, we have that $k + 1 \leq F_t$. Hence $D_{k+1} \leq t$ while $D_{n-k} \geq t$, so

$$B_n(k+1) - B_n(k) < 0.$$

- *Case 3:* $F_{t-1} \leq k \leq F_t - 1$ *and* $F_{t-2} < n - k \leq F_{t-1}$. Then $D_{k+1} = t$ while $D_{n-k} = t - 1$, so
$$B_n(k+1) - B_n(k) = 0.$$

- *Case 4:* $k \geq F_t$. Since $n \leq F_{t+1}$, we have $n - k \leq F_{t-1}$. Therefore $D_{k+1} \geq t + 1$ and $D_{n-k} \leq t - 1$, so
$$B_n(k+1) - B_n(k) > 0.$$

- *Case 5:* $n - k \leq F_{t-2}$. Since $n > F_t$, we have $k + 1 > F_{t-1}$. Hence $D_{k+1} \geq t$ and $D_{n-k} \leq t - 2$, so
$$B_n(k+1) - B_n(k) > 0.$$

Thus, when varying k, the value of $B_n(k)$ plateaus exactly across the values such that
$$F_{t-1} \leq k \leq F_t \quad \text{and} \quad F_{t-2} \leq n - k \leq F_{t-1},$$
with values outside that range resulting in greater $B_n(k)$. □